CS341 Task Tracker – UML and Data Diagrams

English | CS341 Project | Canonical editable sources: docs/plantuml/*.puml

1. Diagram Source and Status

All seven authoritative diagram definitions are provided as PlantUML `.puml` files. Each was rendered with PlantUML 1.2025.10 to both SVG and PNG under `docs/diagrams/`. The figures below use these genuine PlantUML PNG renders; the SVG files are also retained for scalable viewing.

2. Use-case Diagram

The local application distinguishes visitors, signed-in users, task creators and task assignees. Editing and deletion are restricted to task creators.

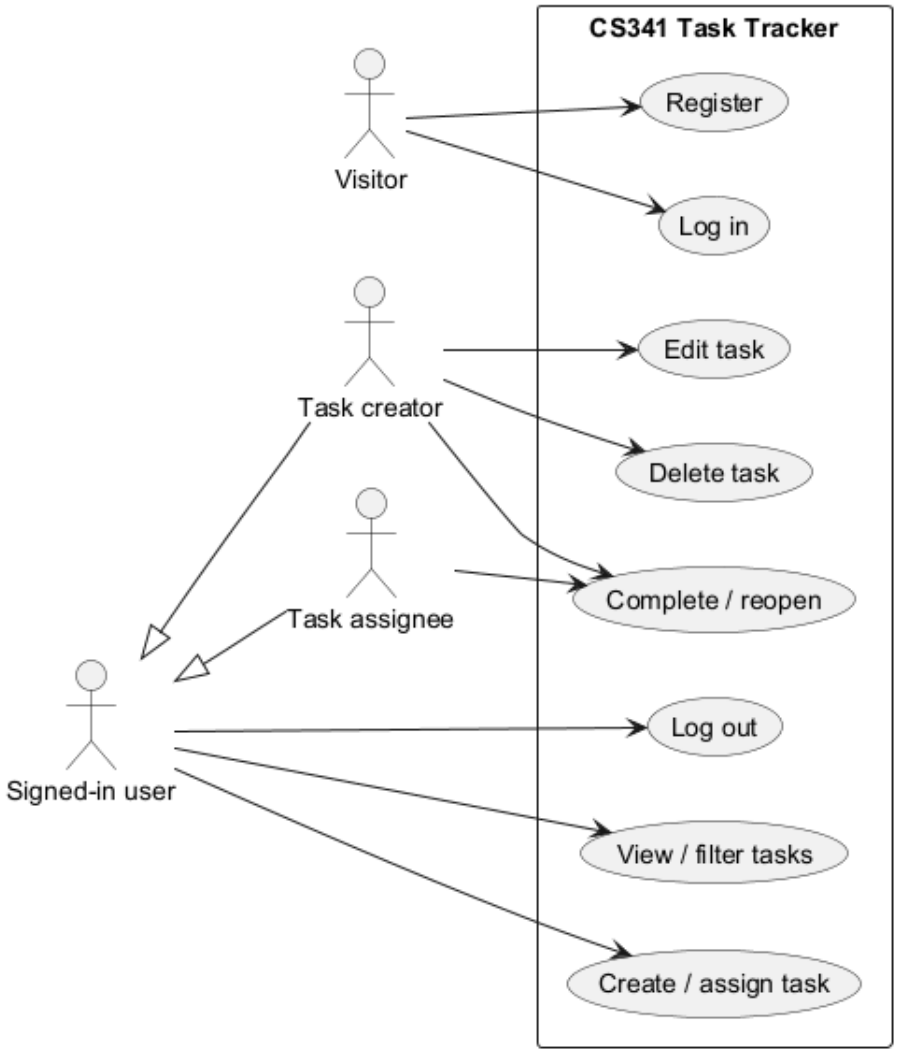


Figure 1: PlantUML render; source: use_case.puml.

3. Class Diagram

View, controller, services, repositories, models and database initialization have separate responsibilities.

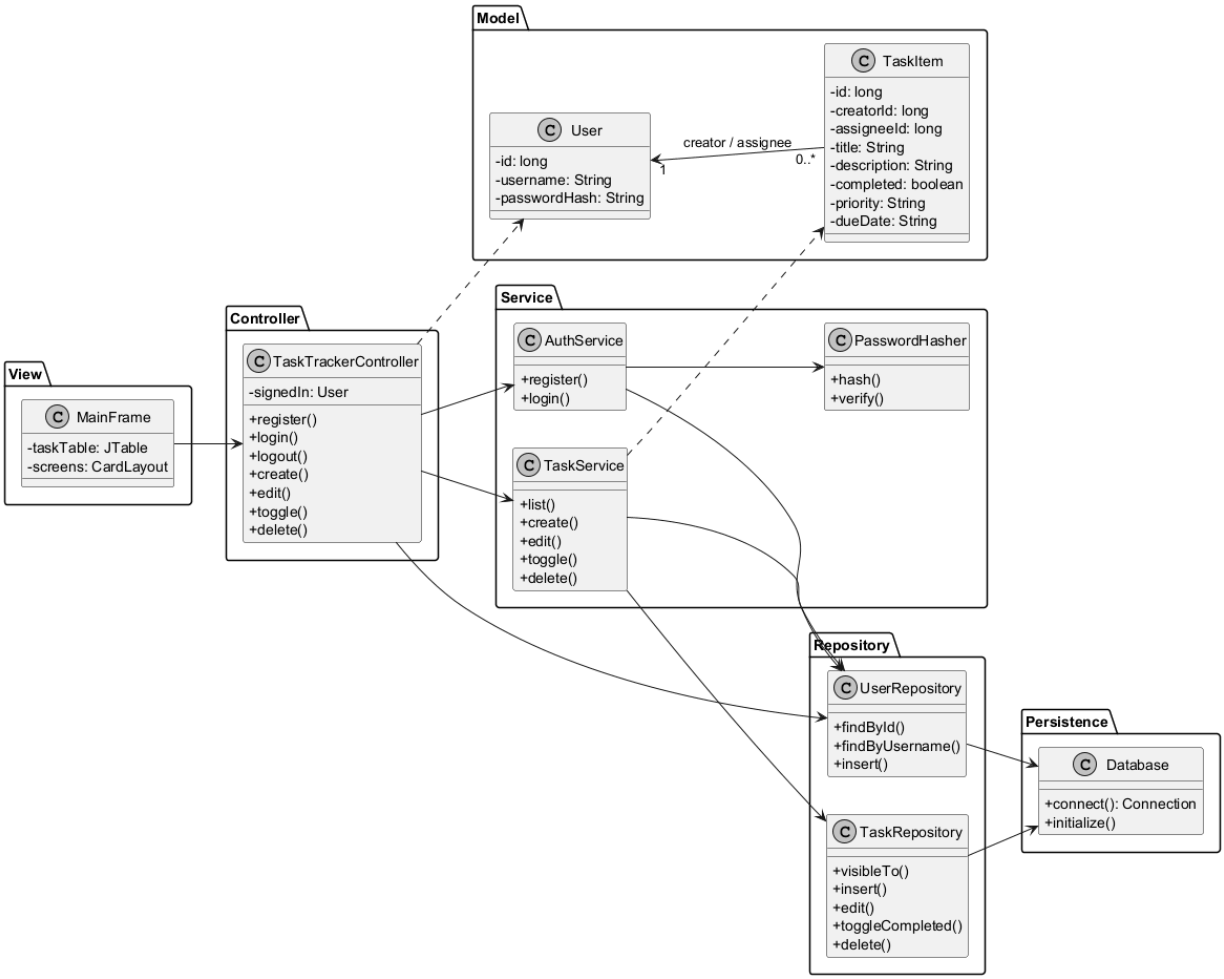


Figure 2: PlantUML render; source: class.puml.

4. Entity Relationship Diagram

Both task creator and assignee are foreign keys to the user table.

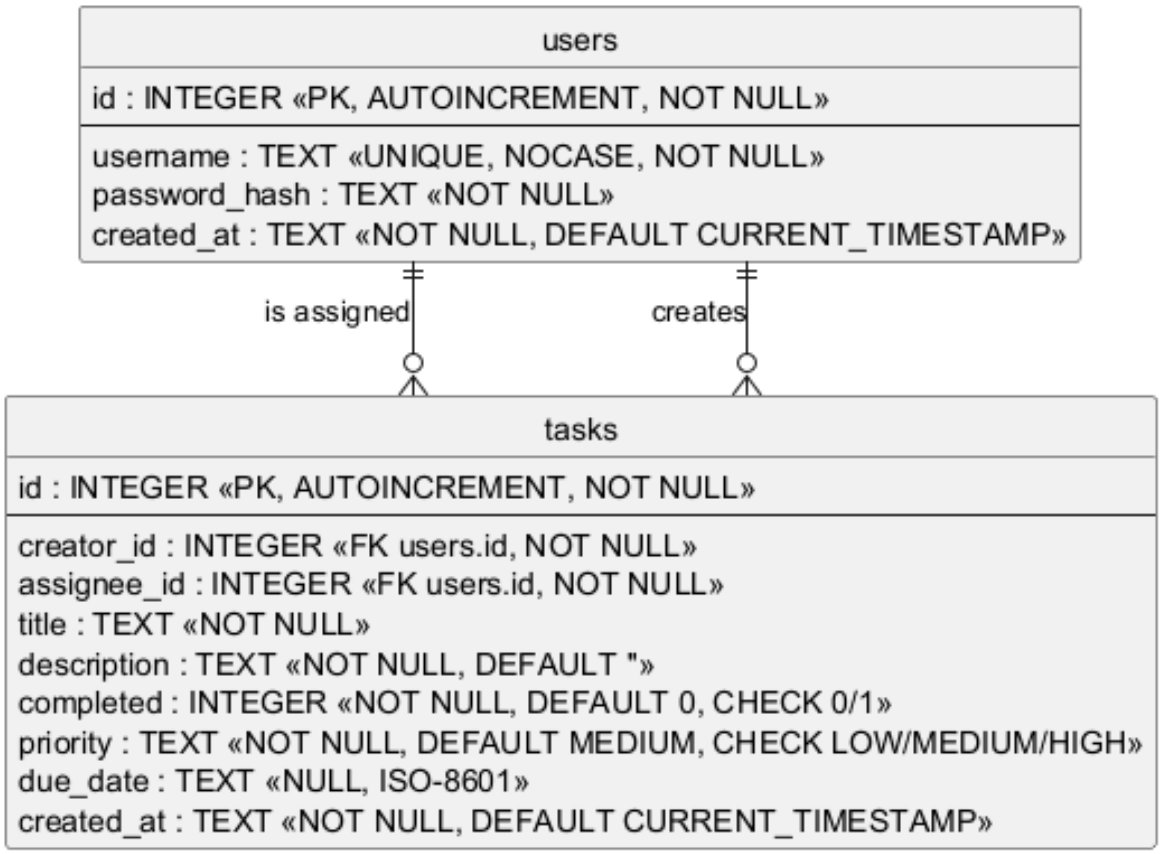


Figure 3: PlantUML render; source: er.puml.

5. Component Diagram

The Swing UI delegates through the controller and services to repositories and the local SQLite database.

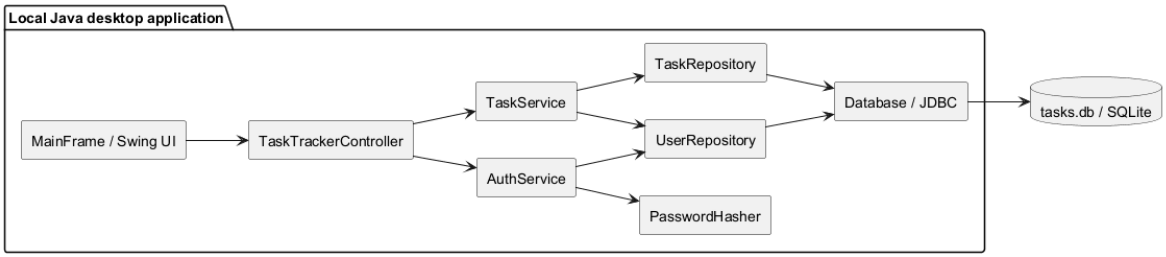


Figure 4: PlantUML render; source: component.puml.

6. Sequence Diagram

Completing a task verifies permission and uses a conditional SQL update.

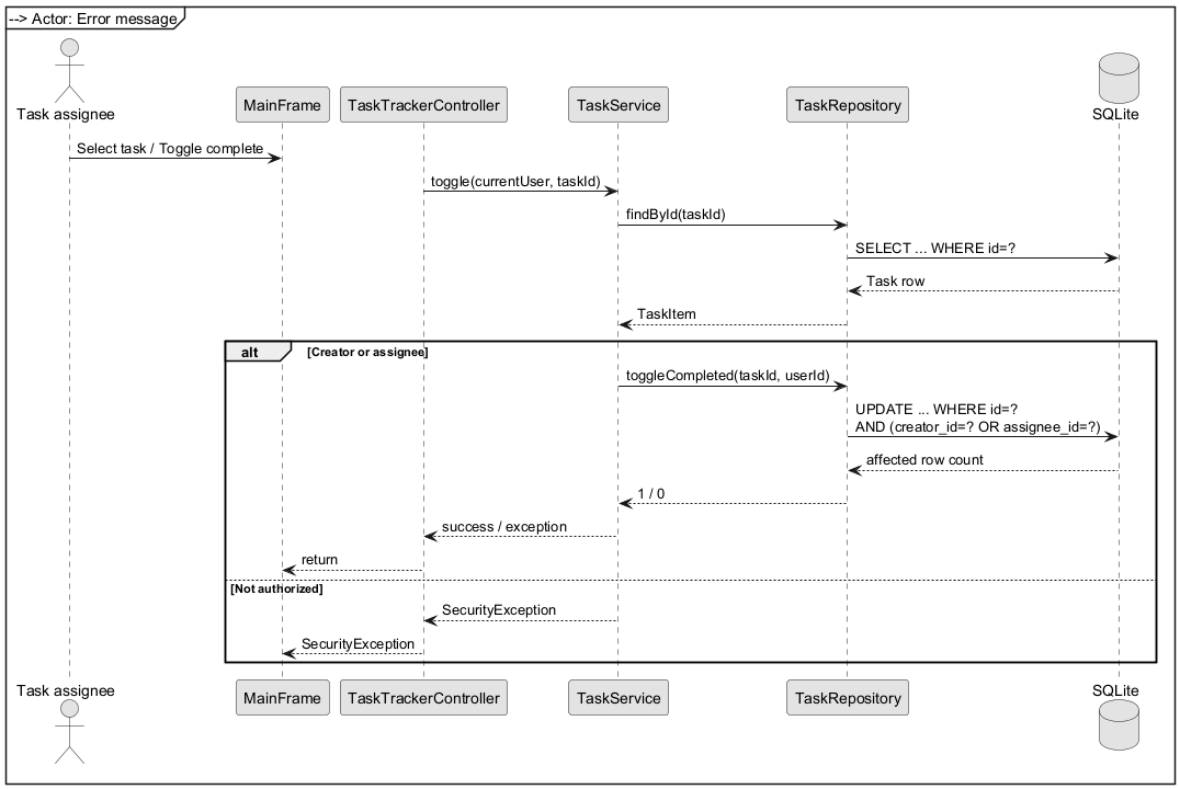


Figure 5: PlantUML render; source: sequence.puml.

7. Activity Diagram

A logged-in user creates, modifies or toggles a task; operations refresh the visible task list.

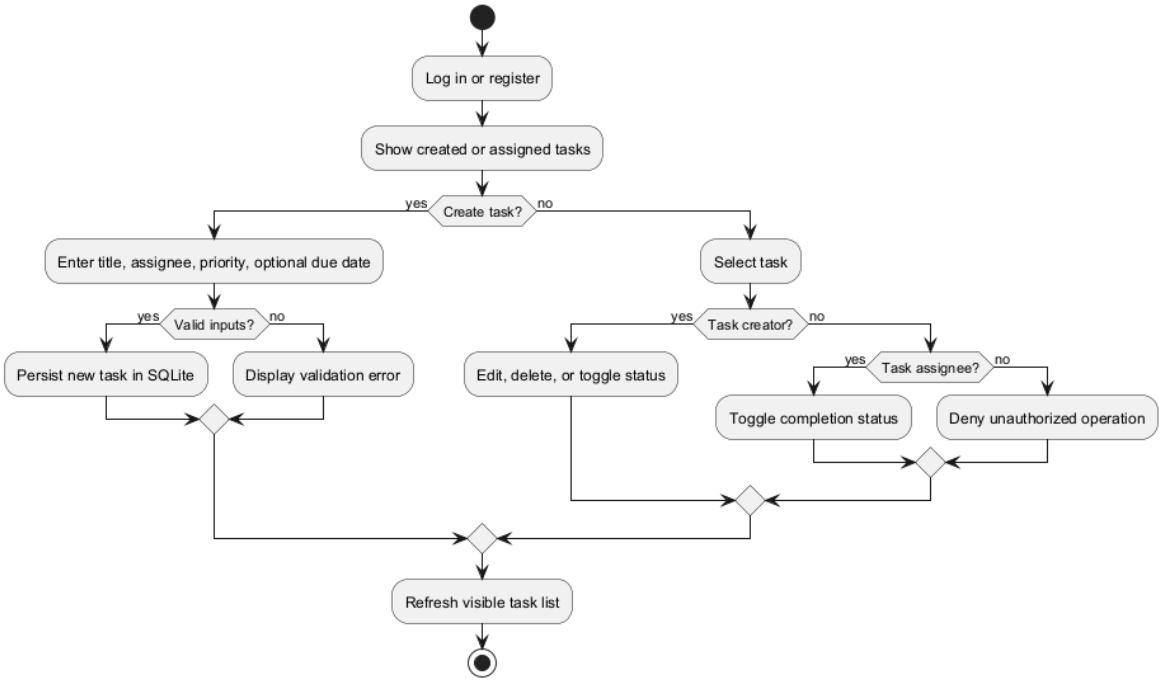


Figure 6: PlantUML render; source: activity.puml.

8. Deployment Diagram

The packaged JAR and local SQLite file reside on one desktop machine, without a required database server.

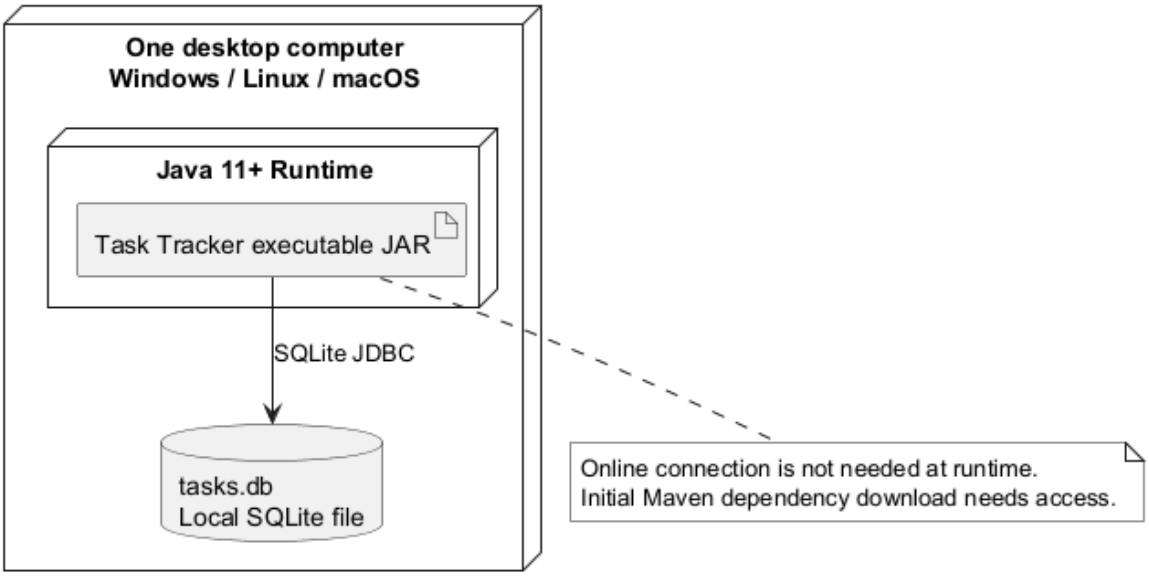


Figure 7: PlantUML render; source: deployment.puml.